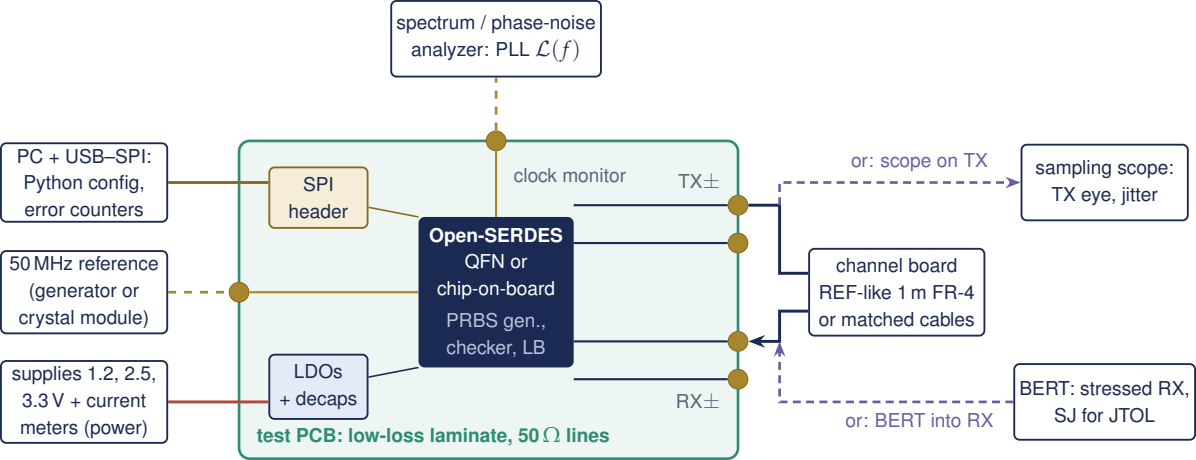
spectrum / phase-noise analyzer: PLL $\mathcal{L}(f)$
PC + USB–SPI: Python config, error counters
SPI header
clock monitor
TX±
Open-SERDES
QFN or chip-on-board
PRBS gen., checker, LB
50 MHz reference (generator or crystal module)
supplies 1.2, 2.5, 3.3 V + current meters (power)
LDOs + decaps
RX±
test PCB: low-loss laminate, 50 Ω lines
or: scope on TX
sampling scope: TX eye, jitter
channel board REF-like 1 m FR-4 or matched cables
BERT: stressed RX, SJ for JTOL
or: BERT into RX